

BVAR vs. Information Criteria: A Monte Carlo Simulation Study

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Abstract

This is the abstract for your paper. Briefly state the purpose of the research, the principal results, and major conclusions. The abstract should be self-contained and citation-free.

Keywords: Keyword 1, Keyword 2, Keyword 3

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1 Introduction

2 Methodology

Describe your structural VAR (SVAR) models and the Monte Carlo Data Generating Processes (DGPs) here.

2.1 Data Generating Process

Explain the true DGP based on Johansen and Juselius (1990), Ramey (2011), or Kilian and Murphy (2014).

3 Simulation Results

Present the results of your Monte Carlo simulations.

4 Conclusion

Summarize the main findings and implications for applied macroeconomic research.

Table 1: Relative Mean Squared Error: Hard Selection (ICs) vs. Bayesian Shrinkage. Bold values indicate the best performing estimator for a given Sample Size.

p_0	Estimator	T	AIC	AICc	SIC (BIC)	HQC	BVAR ($\tau = 0.05$)	BVAR ($\tau = 0.20$)	BVAR ($\tau = 0.50$)	BVAR-C ($\tau = 0.20$)
4	96		1.775	1.025	1.042	0.955	3.222	2.023	1.424	2.057
	144		0.966	1.026	0.992	0.958	2.341	1.334	1.069	1.365
	240		1.032	1.183	1.014	1.004	2.165	1.326	1.079	1.231
	480		1.037	1.366	1.174	1.191	2.472	1.347	1.070	1.430
6	96		1.851	0.940	0.915	0.881	3.424	1.937	1.223	1.972
	144		0.998	1.081	1.066	1.044	2.386	1.267	0.980	1.291
	240		1.189	1.278	1.259	1.308	2.466	1.202	0.963	1.131
	480		1.005	1.384	1.391	1.301	1.986	0.974	0.866	0.997
8	96		1.438	0.793	0.787	0.776	2.703	1.681	1.066	1.616
	144		0.980	0.883	0.887	0.909	2.136	1.167	0.874	1.141
	240		0.942	0.974	0.919	0.901	1.955	1.105	0.857	1.077
	480		1.001	1.218	1.017	0.961	1.976	1.052	0.909	1.017
10	96		1.210	0.561	0.574	0.619	1.757	1.038	0.732	1.044
	144		0.762	0.694	0.685	0.669	1.302	0.758	0.630	0.727
	240		0.830	0.759	0.767	0.766	1.263	0.695	0.654	0.745
	480		0.854	0.916	0.876	0.887	1.296	0.700	0.714	0.758

Table 2: Relative Mean Squared Error: Hard Selection (ICs) vs. Bayesian Model Averaging (BMA). Bold values indicate the best performing estimator for a given Sample Size.

p_0	Estimator	AIC	AICc	SIC (BIC)	HQC	OLS BMA (BIC-W)	BVAR-BIC	BVAR-BMA
T								
4	96	1.775	1.025	1.042	0.955	1.017	0.995	0.895
	144	0.966	1.026	0.992	0.958	1.068	1.100	1.026
	240	1.032	1.183	1.014	1.004	1.060	1.152	1.179
	480	1.037	1.366	1.174	1.191	1.069	1.528	1.190
6	96	1.851	0.940	0.915	0.881	0.919	0.868	0.885
	144	0.998	1.081	1.066	1.044	1.014	1.011	0.920
	240	1.189	1.278	1.259	1.308	1.155	1.174	1.192
	480	1.005	1.384	1.391	1.301	1.216	1.260	0.979
8	96	1.438	0.793	0.787	0.776	0.739	0.790	0.779
	144	0.980	0.883	0.887	0.909	0.908	0.923	0.930
	240	0.942	0.974	0.919	0.901	1.004	1.140	1.058
	480	1.001	1.218	1.017	0.961	0.893	1.075	0.992
10	96	1.210	0.561	0.574	0.619	0.559	0.492	0.475
	144	0.762	0.694	0.685	0.669	0.747	0.593	0.595
	240	0.830	0.759	0.767	0.766	0.777	0.729	0.648
	480	0.854	0.916	0.876	0.887	0.907	0.788	0.726